NUMERICAL METHODS FOR PDE
Deliverable 2: Adaptivity, transient problems and Navier-Stokes

1. Consider the 1D boundary value problem

$$-u'' = s \quad x \in (0,1), \quad u(0) = u(1) = 0$$

solved in the file `main_FEM_1D_exampleForEE.py` with the auxiliary functions in `femFunctions1D.py`.

   (a) Implement the computation of the *gradient smoothing* q of the FE solution, that is, q is the $\mathcal{L}_2$-projection of du^h/dx into the FE space. Plot in the same figure, the derivative du^h/dx and its continuous approximation q for a mesh with 10 elements.

   (b) Compute the Zienkiewicz and Zhu (ZZ) error estimation in each element. That is,

$$|u^h - u|^2_{\mathcal{H}_1(\Omega_e)} \simeq \int_{\Omega_e} \left(\frac{du^h}{dx} - q\right)^2 dx$$

   and the corresponding estimate of the density of the square error

$$\rho_e := \frac{1}{|\Omega_e|}|u^h - u|^2_{\mathcal{H}_1(\Omega_e)}.$$

   Plot the error map for a mesh with 10 elements.

   (c) Check that $\rho_e \leq \tau$ implies $\rho := \frac{1}{|\Omega|}|u^h - u|^2_{\mathcal{H}_1(\Omega)} \leq \tau$.

   (d) Use the function `refineMesh1DLinear` to split in two elements the elements with $\rho_e > \tau$ with $\tau = 0.1$.

   (e) Repeat the computations with the new mesh. Repeat the whole process at least 4 times, showing the error map and the plots you consider that are interesting, in each iteration.

2. Consider the following problem statement modeling, for instance, the heating of a thin film of certain material which was initially at room temperature (16°C), when it is introduced into a an oven:

$$\begin{aligned} \frac{\partial u}{\partial t} &= \boldsymbol{\Delta} u, \quad \mathbf{x} \in \Omega = (0,5)^2, \ t > 0, \\ u(\mathbf{x}, t) &= g(t), \quad \forall (\mathbf{x}, t) \in \partial\Omega \times (0, +\infty), \\ u(\mathbf{x}, 0) &= 16, \quad \forall \mathbf{x} \in \Omega, \end{aligned}$$

   with

$$g(t) = -\frac{300}{t+3} + 116 \ ^\circ\text{C}$$

   representing the heating of the walls of the thin film as the time evolves.

   (a) Derive the weak formulation associated to the problem. State the initial value problem resulting of the discretisation of the weak formulation using the approximation $u^h(\mathbf{x}, t) = \sum_{i=1}^n u_i(t) N_i(\mathbf{x})$, with N_i the basis functions satisfying $N_i(\mathbf{x}_j) = \delta_{ij}$.

   (b) Assuming that the eigenvalues of the matrix $M^{-1}K$, $\{\lambda_i\}_{i=1}^N$, satisfy $0 \leq \lambda_i \leq \lambda$ with $\lambda = \alpha/h^2$, $\alpha > 0$, state the stability condition for the Euler explicit and Euler implicit time integration schemes.

   (c) Implement the solution of the above stated system of ODE using linear, quadrilateral (Q1) elements and Euler explicit time integration (with lumped mass matrix). At which time, T, will the walls of the thin film be at 66°C? Provide an accurate plot of the evolution of the temperature u at $\mathbf{x} = (2.5, 2.5)$ for $t \in [0, T]$.

   (d) Implement Euler implicit time integration. How much time does it take to achieve a higher temperature than 85°C in the whole film? Give the time approximately.

3. Solve the cavity flow problem with the highest possible Reynolds number. Explain the method and the incremental strategy used for the solution of the non-linear problem. Show a plot of the solution in each incremental step.

To upload in Atenea:

- A pdf file `H2_2425_SurnameName.pdf` containing a self-contained report with the answers to the questions of the assignment.

- A zip file `H2_2425_CodesSurnameName.zip` containing the .py files and all necessary Python files for running the computations.